EX: 4	Practice the Nmap in Kali Linux
DATE:	

AIM:

To perform a network scan using Nmap, identify active hosts, open ports, analyze the results, and understand potential vulnerabilities.

PROCEDURE:

STEP 1: Install Nmap

- **sudo apt update && sudo apt install nmap**

```
Shell No. 1
Session  Actions  Edit  View  Help

kali ~  20:20 ❯ sudo apt update && sudo apt install nmap
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:2 https://repo.protonvpn.com/debian stable InRelease
Hit:3 http://http.kali.org/kali kali-rolling InRelease
881 packages can be upgraded. Run 'apt list --upgradable' to see them.
nmap is already the newest version (7.98+dfsg-1kali1).
The following packages were automatically installed and are no longer required:
  curlftpfs      libconfig-inifiles-perl  libmpeg2encpp-2.1-0t64  libpostproc58      libvdpau-va-gl1     vdpau-driver-all
  libavfilter10  libgav1-1                libmplex2-2.1-0t64      libsphinxbase3t64  mesa-vdpau-drivers
  libavformat61  libmjpegutils-2.1-0t64   libpocketsphinx3        libswscale8        pocketsphinx-en-us
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 881

kali ~  20:20 ❯
```

STEP 2: Identify the Target Network

- **ip a**

```
Shell No. 1
Session  Actions  Edit  View  Help

kali ~  20:21 ❯ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:08:60:27 brd ff:ff:ff:ff:ff:ff
    inet 192.168.127.128/24 brd 192.168.127.255 scope global dynamic noprefixroute eth0
       valid_lft 1648sec preferred_lft 1648sec
    inet6 fe80::4f1c:9edf:98c4:c967/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

kali ~  20:21 ❯
```

STEP 3: Scan the Network

- **Discover Active Hosts** (Ping Scan):
  - **nmap -sn 192.168.1.0/24**

```
kali ~ 20:27 ❯ nmap -sn 192.168.127.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:27 +0530
Nmap scan report for 192.168.127.1
Host is up (0.00065s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.127.2
Host is up (0.00020s latency).
MAC Address: 00:50:56:FA:5C:4C (VMware)
Nmap scan report for 192.168.127.254
Host is up (0.00018s latency).
MAC Address: 00:50:56:F6:D3:75 (VMware)
Nmap scan report for 192.168.127.128
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.93 seconds
```

- **For specific host:**
  - **nmap 192.168.1.0**

```
kali ~ 20:29 ❯ nmap -sn 192.168.127.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:29 +0530
Nmap scan report for 192.168.127.1
Host is up (0.00052s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds
```

STEP 4: Scan all Ports: nmap -p- 192.168.1.0

```
kali ~ 20:42 ❯ nmap -p- 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:42 +0530
Stats: 0:01:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 20:44 (0:00:00 remaining)
Nmap scan report for 192.168.1.1
Host is up (0.070s latency).
Not shown: 65527 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
23/tcp    filtered telnet
53/tcp    open     domain
80/tcp    open     http
443/tcp   open     https
7547/tcp  open     cwmp
18182/tcp open     opsec-ufp
52881/tcp open     unknown
MAC Address: B8:B7:DB:B1:24:E4 (Goip Global Services Pvt.)

Nmap done: 1 IP address (1 host up) scanned in 110.05 seconds
```

STEP 5: Advanced Scans:

- **Operating System Detection:**

  nmap -O 192.168.1.1

```
Session  Actions  Edit  View  Help

kali ~  20:50  in 1m50s122ms ❯ nmap -O 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:50 +0530
Nmap scan report for 192.168.1.1
Host is up (0.021s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE
22/tcp   open     ssh
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
MAC Address: B8:B7:DB:B1:24:E4 (Goip Global Services Pvt.)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.8 - 2.6.30
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.21 seconds

kali ~  20:50  in 4s287ms ❯
```

- **Perform Service and Version Detection:**

  nmap -sV 192.168.1.1

```
kali ~  20:50  in 4s287ms ❯ nmap -sV 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:53 +0530
Nmap scan report for 192.168.1.1
Host is up (0.17s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE    VERSION
22/tcp   open     ssh        Dropbear sshd 0.48 (protocol 2.0)
23/tcp   filtered telnet
53/tcp   open     domain     dnsmasq 2.80
80/tcp   open     tcpwrapped
443/tcp  open     tcpwrapped
MAC Address: B8:B7:DB:B1:24:E4 (Goip Global Services Pvt.)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.80 seconds

kali ~  20:54  in 15s867ms ❯
```

- **Vulnerability Scanning:**
  nmap --script vuln 192.168.1.1

```
Session  Actions  Edit  View  Help
kali ~  20:58  in 1m56s108ms ❯ nmap --script vuln 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 20:58 +0530
Nmap scan report for 192.168.1.1
Host is up (0.041s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE
22/tcp   open     ssh
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
| http-vuln-cve2010-0738:
|_  /jmx-console/: Authentication was not required
|_http-majordomo2-dir-traversal: ERROR: Script execution failed (use -d to debug)
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
| http-enum:
|_  /admin/login.asp: Possible admin folder
443/tcp open     https
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-aspnet-debug: ERROR: Script execution failed (use -d to debug)
| ssl-heartbleed:
|   VULNERABLE:
|   The Heartbleed Bug is a serious vulnerability in the popular OpenSSL cryptographic software library. It allows for stealing information intended to be pr
otected by SSL/TLS encryption.
|     State: VULNERABLE
|     Risk factor: High
|       OpenSSL versions 1.0.1 and 1.0.2-beta releases (including 1.0.1f and 1.0.2-beta1) of OpenSSL are affected by the Heartbleed bug. The bug allows for r
eading memory of systems protected by the vulnerable OpenSSL versions and could allow for disclosure of otherwise encrypted confidential information as well
as the encryption keys themselves.
|
|     References:
|       http://cvedetails.com/cve/2014-0160/
|       http://www.openssl.org/news/secadv_20140407.txt
|_      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-0160
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
|_http-csrf: Couldn't find any CSRF vulnerabilities.
| ssl-poodle:
|   VULNERABLE:
|   SSL POODLE information leak
|     State: VULNERABLE
|     IDs:  BID:70574  CVE:CVE-2014-3566
|           The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
|           products, uses nondeterministic CBC padding, which makes it easier
|           for man-in-the-middle attackers to obtain cleartext data via a
|           padding-oracle attack, aka the "POODLE" issue.
```

STEP 6: Mitigate Vulnerabilities

- **Close unused ports:**
  sudo service <service> stop

```
kali ~  21:13 ❯ sudo service ssh stop

kali ~  21:13 ❯ nmap 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-03 21:13 +0530
Nmap scan report for 192.168.1.1
Host is up (0.056s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE
23/tcp   filtered telnet
53/tcp   open     domain
80/tcp   open     http
80/tcp   open     https
443/tcp  open     https
MAC Address: B8:B7:DB:B1:24:E4 (Goip Global Services Pvt.)

Nmap done: 1 IP address (1 host up) scanned in 4.36 seconds

kali ~  21:13  in 4s420ms ❯
```

Result:
Basic Nmap commands were successfully practiced and understood